

Quiz 2 - Part 2: Discrete Mathematics (CSE121)

Date: 15-09-2025

Maximum marks: 30

Instructions. Please read the following points carefully.

- Mention your names, roll numbers, room number, and the part number of the quiz in your answer sheets. If we find any details missing, we will not grade the answer sheet.
- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- If you are found copying, the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or TAs.
- This exam is designed for 50 minutes.

1 Problem Set

Question 1.1. *Prove the following only using the properties of integers (given at the end). You should mention which rule you are using at every step.*

1. (2 marks). If $1 = -1$ then prove that $2 = 4$.

Answer + Rubric. Given $1 = -1$. Add 1 to both sides. Then, we get

$$1 + 1 = -1 + 1$$

(It was shown in the third tutorial that using the 12 properties of integers, we can do the following: for every integers a, b, c , if $a = b$ then $a + c = b + c$. If you mentioned it but not proved, no mark will be deducted. If you did not mention this, 0.5 marks will be deducted.) Now, use the property of additive inverse to get $-1 + 1 = 0$. If it is not mentioned, 0.5 marks will be deducted. So, we have

$$2 = 0.$$

Now, add 2 to both sides. If this is not mentioned, 0.5 marks will be deducted. We get

$$2 = 4.$$

If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

2. (4 marks). Is the following true: if $1 = -1$ then $2 = 1$? Prove your answer.

Answer + Rubric. Unfortunately, I could not find a correct proof! I thought the second incorrect method given below is correct, but there was a bug. I took this question from 'Mathematics for Computer Science' and the authors also made the same mistake. Sorry about that! Everyone will get full marks in this question. I am presenting two incorrect proofs here, which are very subtle and appear almost correct. I request you to kindly go through these.

Incorrect proof 1. Given $1 = -1$. By the first part, we know that $2 = 0$. Now, add 2 to both sides of $2 = 0$. Then,

$$4 = 2.$$

Now, add the additive inverse of 2 to both sides. We get

$$4 - 2 = 0.$$

On using distributivity, we get

$$2(2 - 1) = 0.$$

By no zero divisors property, we get either $2 = 0$ or $2 - 1 = 0$. As mentioned in the first part, if $1 = -1$ then $2 = 0$. Then, in this case, we may not have $2 - 1 = 0$. This is the reason why this proof is incorrect, however the argument is very subtle. If someone has written this, they will get 3.5 marks out of 4, provided all the properties mentioned above are used.

Incorrect proof 2. One way to do this is to treat integers as rational number and then working with rational numbers.

We are given $1 = -1$, which is the same as

$$\frac{1}{1} = \frac{-1}{1}.$$

On multiplying with $\frac{1}{2}$, we get

$$\frac{1}{2} = \frac{-1}{2}.$$

Add $\frac{3}{2}$ to the equation above. Then, we get

$$\frac{1}{2} + \frac{3}{2} = \frac{-1}{2} + \frac{3}{2}$$

It can be shown using the 12 properties of integers that for every rational number $\frac{a}{b}, \frac{c}{d}, \frac{e}{f}$, if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a}{b} + \frac{e}{f} = \frac{c}{d} + \frac{e}{f}$ and $\frac{a}{b} \cdot \frac{e}{f} = \frac{c}{d} \cdot \frac{e}{f}$. If you don't mention this, no marks will be deducted. On simplifying the equation above,

$$\frac{2}{1} = \frac{1}{1}.$$

As $\frac{2}{1}, \frac{1}{1}$ are integers, they can be written as 2, 1 and hence, we get $2 = 1$.

The problem with this proof is that if $1 = -1$, we get $2 = 0$. Thus, $\frac{1}{2}$ is actually $\frac{1}{0}$, which is undefined! Hence, this proof is incorrect.

3. (4 marks). Let a, b, c be integers. If $a \mid b$ and $b \mid c$ then prove that $a^2 \mid c^2$.

Answer + Rubric. We are given that $a \mid b$ and $b \mid c$. This means there exist integers m, n such that $b = a \cdot m$ and $c = b \cdot n$. If this is not mentioned, 0.5 marks should be deducted.

Now, $b^2 = a^2 m^2$ and $c^2 = b^2 n^2$ (by using commutativity of multiplication). If this is not mentioned, 1.5 marks should be deducted. Substitute $b^2 = a^2 m^2$ in $c^2 = b^2 n^2$. We get $c^2 = a^2 m^2 n^2$. If this is not mentioned, deduct 0.5 marks. Thus, $a^2 \mid c^2$.

If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Question 1.2. Prove the following. You are only allowed to use the properties of integers given at the end of this document. You should mention which rule you are using at every step.

1. (4 marks). For two integers a, b , $\max(a, b)$ and $\min(a, b)$ denote the maximum and the minimum of two numbers, respectively and if $a = b$ then $\max(a, b) = \min(a, b)$.

Prove that for every integers a, b , where a, b can also be the same,

$$\min(a, b) + \max(a, b) = a + b.$$

Answer + Rubric. This is proved by cases. Given a, b , there are two possibilities. Either $a \leq b$ or $a > b$. If this is not mentioned, 1 point should be deducted. When $a \leq b$, then $\min(a, b) = a$ and $\max(a, b) = b$. If this is not mentioned, 1 mark should be deducted. Hence, $\max(a, b) + \min(a, b) = a + b$. Otherwise, $a > b$, in which case $\min(a, b) = b$ and $\max(a, b) = a$. If this is not mentioned, 1 mark should be deducted. Hence, we get $\max(a, b) + \min(a, b) = a + b$.

If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

2. (6 marks). Let a, b be integers. Prove that if both a^2 and b^2 are even, then $a + b$ and $a \cdot b$ are also even.

Answer+Rubric. Given both a^2 and b^2 are even. Then, we have already done in class that this implies a, b are even. However, you should have proven this. If this proof is missing, 3 marks will be deducted. Now, there is a straight forward direct proof which shows that $a + b$ and $a \cdot b$ are even. If one of this is not mentioned, 1 mark will be deducted, otherwise 2 marks will be deducted.

If it is proved using any other way, then every step of that proof must follow from the 12 properties of integers and appropriate property should be mentioned at every step. The marks will be assigned accordingly.

Question 1.3. 1. (5 marks). Prove that $\sqrt{2}$ is irrational.

Answer + Rubric. This was already done in the 10th lecture. The marks will be given as per the proof discussed in the class.

2. (2 marks). Assume that the following statements are true. What will be the conclusion of the following statements? Justify your answer.

- (a) If a student attends lectures, then they learn concepts.
- (b) If they learn concepts, then they pass the exam.
- (c) Ram attends lectures.

Answer + Rubric. This is proved using Modus Ponens. Let p, q, r, s be the propositions ‘a student attends lectures’, ‘students learn concepts’, ‘students pass the exam’, and ‘Ram attends lectures’. We are given that $p \implies q, q \implies r, s$ are true. Then by Modus Ponens, we first get that Ram learns concepts and then we get that Ram passes the exam. One mark will be given to mention Modus Ponens and 0.5 marks will be given to each of the two conclusions we get.

- 3. (3 marks). Prove that an integer n is even if and only if $n + 1$ is odd. You are only allowed to use the properties of integers given below.

Answer + Rubric. There are two parts here: if n is even then $n + 1$ is odd and if $n + 1$ is odd then n is even. The proofs of these parts are straightforward and use the definitions of odd and even integers. 1.5 marks will be given to every part. One mark will be deducted for not using the definitions of odd and even integers.

A Important properties of integers

The following properties hold for every integers a, b, c .

1. **Closure under addition and multiplication:** $a + b, a \cdot b$ are integers.
2. **Associativity of addition:** $a + (b + c) = (a + b) + c$.
3. **Associativity of multiplication:** $a \cdot (b \cdot c) = (a \cdot b) \cdot c$.
4. **Commutativity of addition:** $a + b = b + a$.
5. **Commutativity of multiplication:** $a \cdot b = b \cdot a$.
6. **Distributive properties:** $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.
7. **Identities:** $0 \neq 1, a + 0 = a, a \cdot 0 = 0$, and $a \cdot 1 = a$.
8. **Additive inverses:** There exists a unique integer m such that $m := -1 \cdot a$ and $a + m = 0$. The inverse of a is denoted $-a$.
9. **Subtraction:** $a - b$ is defined as $a + (-b)$.
10. **Cancellation:** If $a \cdot b = a \cdot c$ and $a \neq 0$ then $b = c$.
11. **No zero divisors:** If $a \cdot b = 0$ then either $a = 0$ or $b = 0$.
12. **Signed product:** $-a \cdot b = (-a) \cdot b = a \cdot (-b)$.